

Bitwise linear trail through **ARADI** found by SAT

CiVerLy

July 8, 2026

1 ARADI

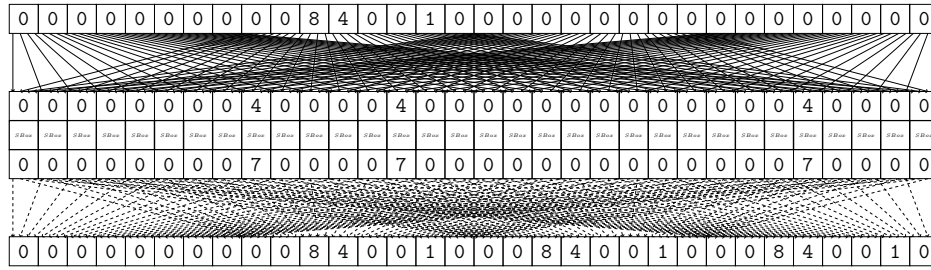
Maximal linear correlation: 2^{-7} [illegible]

2 ARADI-round0

Maximal linear correlation: $2^{-3.0}$

0	0	0	0	0	0	0	0	0	0	0	8	4	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	8	4	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
<i>BN</i>																																	
0	0	0	0	0	0	0	0	0	0	0	8	4	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	8	4	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
<i>SBiasLayer</i>																																	
0	0	0	0	0	0	0	0	0	0	0	8	4	0	0	1	0	0	0	8	4	0	0	1	0	0	0	8	4	0	0	1	0	0
0	0	0	0	0	0	0	0	0	0	0	8	4	0	0	1	0	0	0	8	4	0	0	1	0	0	0	8	4	0	0	1	0	0
<i>LinearLayer0</i>																																	
0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	4	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	4	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	4	0	0	0	0	0	0

3 SBoxLayer

Maximal linear correlation: $2^{-3.0}$ 

4 LinearLayer0

Maximal linear correlation: $2^{-0.0}$

0	0	0	0	0	0	0	0	0	0	0	8	4	0	0	1	0	0	0	8	4	0	0	1	0	0	0	8	4	0	0	1	0							
0	0	0	0	0	0	0	0	0	0	0	8	4	0	0	1	0	0	0	8	4	0	0	1	0	0	0	8	4	0	0	1	0							
L0										L0										L0										L0									
0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	4	0	0	0	0							
0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	4	0	0	0	0							

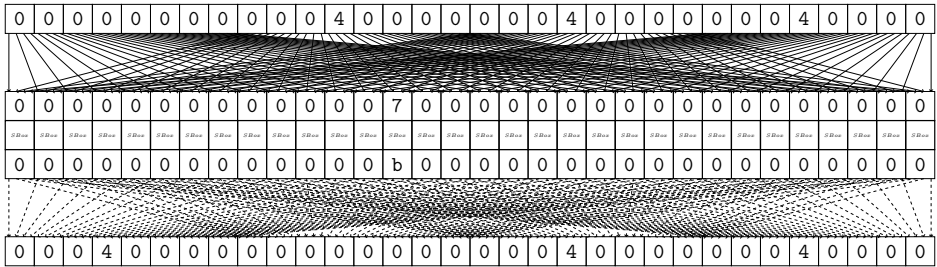
5 ARADI-round1

Maximal linear correlation: $2^{-1.0}$

0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	4	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	4	0	0	0
BN																															
0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	4	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	4	0	0	0	
SBiasLayer																															
0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	4	0	0	0	
0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	4	0	0	0	
LinearLayer1																															
0	1	0	4	0	0	8	0	0	0	0	0	0	0	0	0	0	1	0	4	0	0	8	0	0	1	0	4	0	0	8	0
0	1	0	4	0	0	8	0	0	0	0	0	0	0	0	0	0	1	0	4	0	0	8	0	0	1	0	4	0	0	8	0

6 SBoxLayer

Maximal linear correlation: $2^{-1.0}$



7 LinearLayer1

Maximal linear correlation: $2^{-0.0}$

0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	4	0	0	0	0
0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	4	0	0	0	0	
S1								S1								S1								S1								
0	1	0	4	0	0	8	0	0	0	0	0	0	0	0	0	0	1	0	4	0	0	8	0	0	0	1	0	4	0	0	8	0
0	1	0	4	0	0	8	0	0	0	0	0	0	0	0	0	0	1	0	4	0	0	8	0	0	0	1	0	4	0	0	8	0

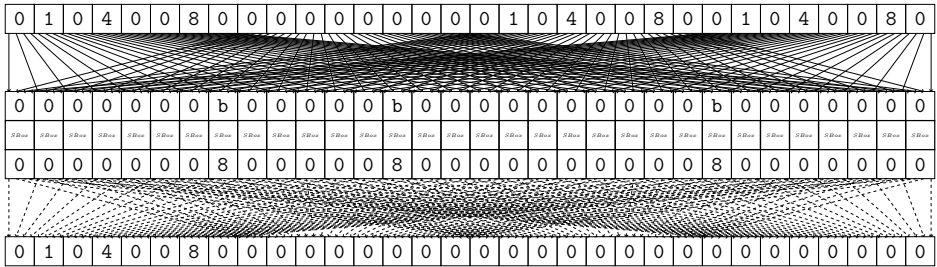
8 ARADI-round2

Maximal linear correlation: $2^{-3.0}$

0	1	0	4	0	0	8	0	0	0	0	0	0	0	0	0	0	1	0	4	0	0	8	0	0	1	0	4	0	0	8	0
0	1	0	4	0	0	8	0	0	0	0	0	0	0	0	0	0	1	0	4	0	0	8	0	0	1	0	4	0	0	8	0
BN																															
0	1	0	4	0	0	8	0	0	0	0	0	0	0	0	0	0	1	0	4	0	0	8	0	0	1	0	4	0	0	8	0
0	1	0	4	0	0	8	0	0	0	0	0	0	0	0	0	0	1	0	4	0	0	8	0	0	1	0	4	0	0	8	0
0	1	0	4	0	0	8	0	0	0	0	0	0	0	0	0	0	1	0	4	0	0	8	0	0	1	0	4	0	0	8	0
SBias Layer																															
0	1	0	4	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	0	4	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	0	4	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Linear Layer 2																															
8	3	0	c	4	4	9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8	3	0	c	4	4	9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

9 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



10 LinearLayer2

Maximal linear correlation: $2^{-0.0}$ [illegible]